

Quiz4 Solutions

1. $\text{logit}[\text{Prob}[Y_i=1]] = \sum_{j=1}^p X_{ij}\beta_j = \tilde{X}_i'\beta$, where $\tilde{X}_i = (X_{i1}, \dots, X_{ip})'$ and $\beta = (\beta_1, \dots, \beta_p)'$.

$$\Rightarrow \Pr(Y_i=1) = \frac{e^{\tilde{X}_i'\beta}}{1+e^{\tilde{X}_i'\beta}}$$

So But if we consider in a Bayesian paradigm and hence we should write.

$$\Pr(Y_i=1|\beta) = \frac{e^{\tilde{X}_i'\beta}}{1+e^{\tilde{X}_i'\beta}} \Rightarrow \Pr(Y_i=y_i|\beta) = \left(\frac{e^{\tilde{X}_i'\beta}}{1+e^{\tilde{X}_i'\beta}}\right)^{y_i} \left(\frac{1}{1+e^{\tilde{X}_i'\beta}}\right)^{1-y_i} = \frac{e^{y_i \tilde{X}_i'\beta}}{1+e^{\tilde{X}_i'\beta}}$$

$$\text{So, } \Pr(Y_1=y_1, \dots, Y_n=y_n|\beta) = \prod_{i=1}^n \Pr(Y_i=y_i|\beta) = \prod_{i=1}^n \frac{e^{y_i \tilde{X}_i'\beta}}{1+e^{\tilde{X}_i'\beta}}$$

$$\pi(\beta) = \pi(\beta_1) \dots \pi(\beta_p) = \prod_{j=1}^p \frac{1}{\sqrt{2\pi}\sigma_0^2} e^{-\frac{1}{2\sigma_0^2} \beta_j^2} \propto e^{-\frac{1}{2\sigma_0^2} \sum_{j=1}^p \beta_j^2}$$

$$\text{So, } \pi(\beta|Y_1=y_1, \dots, Y_n=y_n) \propto \pi(Y_1=y_1, \dots, Y_n=y_n|\beta) \pi(\beta) \\ \propto \prod_{i=1}^n \frac{e^{y_i \tilde{X}_i'\beta}}{1+e^{\tilde{X}_i'\beta}} \times e^{-\frac{1}{2\sigma_0^2} \sum_{j=1}^p \beta_j^2} = \frac{e^{\sum_{i=1}^n y_i \tilde{X}_i'\beta - \frac{1}{2\sigma_0^2} \beta'\beta}}{\prod_{i=1}^n (1+e^{\tilde{X}_i'\beta})}$$

We want the MAP estimator, so, $\hat{\beta}_{\text{MAP}} = \arg\max_{\beta} \pi(\beta|Y_1=y_1, \dots, Y_n=y_n)$
 $= \arg\max_{\beta} \log \pi(\beta|Y_1=y_1, \dots, Y_n=y_n)$.

i.e. we need to solve $\frac{d}{d\beta} \log \pi(\beta|Y_1=y_1, \dots, Y_n=y_n) = 0$.

$$\Rightarrow \frac{d}{d\beta} \left\{ \sum_{i=1}^n y_i \tilde{X}_i'\beta - \frac{1}{2\sigma_0^2} \beta'\beta - \sum_{i=1}^n \log(1+e^{\tilde{X}_i'\beta}) \right\} = 0$$

$$\Rightarrow \sum_{i=1}^n y_i \tilde{X}_i - \frac{1}{2\sigma_0^2} 2\beta - \sum_{i=1}^n \frac{e^{\tilde{X}_i'\beta}}{1+e^{\tilde{X}_i'\beta}} \tilde{X}_i = 0$$

$$\begin{aligned}
 2/ \quad V(Y_{1,1,1}) &= V(\mu + \alpha_1 + \beta_1 + \varepsilon_{1,1,1}) \\
 &= V(\alpha_1) + V(\beta_1) + V(\varepsilon_{1,1,1}) \\
 &= \sigma_\alpha^2 + \sigma_\beta^2 + \sigma_\varepsilon^2
 \end{aligned}$$

$$\begin{aligned}
 V(Y_{2,1,1}) &= V(\mu + \alpha_2 + \beta_1 + \varepsilon_{2,1,1}) \\
 &= V(\alpha_2) + V(\beta_1) + V(\varepsilon_{2,1,1}) \\
 &= \sigma_\alpha^2 + \sigma_\beta^2 + \sigma_\varepsilon^2
 \end{aligned}$$

$$\begin{aligned}
 \text{Cov}(Y_{1,1,1}, Y_{2,1,1}) &= \text{Cov}(\mu + \alpha_1 + \beta_1 + \varepsilon_{1,1,1}, \mu + \alpha_2 + \beta_1 + \varepsilon_{2,1,1}) \\
 &= \text{Cov}(\alpha_1, \alpha_2) + \text{Cov}(\beta_1, \beta_1) + \text{Cov}(\varepsilon_{1,1,1}, \varepsilon_{2,1,1}) \\
 &= \sigma_\beta^2
 \end{aligned}$$

$$\Rightarrow \text{Corr}(Y_{1,1,1}, Y_{2,1,1}) = \sigma_\beta^2 / (\sigma_\alpha^2 + \sigma_\beta^2 + \sigma_\varepsilon^2)$$

$$\begin{aligned}
 V(Y_{1,2,3}) &= V(\mu + \alpha_1 + \beta_2 + \varepsilon_{1,2,3}) = V(\alpha_1) + V(\beta_2) + V(\varepsilon_{1,2,3}) \\
 &= \sigma_\alpha^2 + \sigma_\beta^2 + \sigma_\varepsilon^2
 \end{aligned}$$

$$\begin{aligned}
 V(Y_{2,1,3}) &= V(\mu + \alpha_2 + \beta_1 + \varepsilon_{2,1,3}) = V(\alpha_2) + V(\beta_1) + V(\varepsilon_{2,1,3}) \\
 &= \sigma_\alpha^2 + \sigma_\beta^2 + \sigma_\varepsilon^2
 \end{aligned}$$

$$\begin{aligned}
 \text{Cov}(Y_{1,2,3}, Y_{2,1,3}) &= \text{Cov}(\alpha_1 + \beta_2 + \varepsilon_{1,2,3}, \alpha_2 + \beta_1 + \varepsilon_{2,1,3}) \\
 &= \text{Cov}(\alpha_1, \alpha_2) + \text{Cov}(\beta_2, \beta_1) + \text{Cov}(\varepsilon_{1,2,3}, \varepsilon_{2,1,3}) \\
 &= 0
 \end{aligned}$$

$$\Rightarrow \text{Corr}(Y_{1,2,3}, Y_{2,1,3}) = 0$$

$$3/ \underline{\mu} = \begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix} \sim \text{BVN} \left(\begin{pmatrix} 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \right).$$

$$\underline{X} | \underline{\mu} \sim N_4 \left(\begin{bmatrix} \underline{\mu} \\ \underline{\mu} \end{bmatrix}, \begin{bmatrix} I_2 \\ I_2 \end{bmatrix} \right)$$

This happens due to the inherent structure of R.

$$X \leftarrow c(1, 2, 3), Y \leftarrow c(1, 4, 5, 2, 3, 6)$$

$$X + Y = c(\underbrace{1, 2, 3}_{\text{repeat}}, \underbrace{1, 2, 3}_{\text{repeat}}) + c(1, 4, 5, 2, 3, 6)$$

We can also write it as $\underline{1}_2 \otimes \underline{\mu}$.

$$= c(1+1, 2+4, 3+5, 1+2, 2+3, 3+6)$$

Where $\underline{1}_2 = (1, 1)'$ and $\otimes$ is the Kronecker product.

$$\pi(\underline{X}) = \int \pi(\underline{X} | \underline{\mu}) \pi(\underline{\mu}) d\underline{\mu}$$

$$= \int \frac{1}{|2\pi I_4|^{1/2}} e^{-\frac{1}{2} \left[\underline{X} - \begin{pmatrix} \underline{\mu} \\ \underline{\mu} \end{pmatrix} \right]' \left[\underline{X} - \begin{pmatrix} \underline{\mu} \\ \underline{\mu} \end{pmatrix} \right]} \frac{1}{|2\pi I_2|^{1/2}} \times e^{-\frac{1}{2} (\underline{\mu} - \underline{1})' (\underline{\mu} - \underline{1})} d\underline{\mu}$$

$$\propto \int e^{-\frac{1}{2} \left\{ \begin{bmatrix} X_1 - \mu_1 \\ X_2 - \mu_2 \end{bmatrix}' \begin{bmatrix} X_1 - \mu_1 \\ X_2 - \mu_2 \end{bmatrix} + (\underline{\mu} - \underline{1})' (\underline{\mu} - \underline{1}) \right\}} d\underline{\mu}$$

$$\underline{X} = \begin{pmatrix} X_1 \\ X_2 \\ \sim \\ \sim \end{pmatrix} \begin{cases} \text{first} \\ \text{two} \\ \text{third} \\ \text{and} \\ \text{fourth} \\ \text{element} \end{cases}$$

$$\propto \int e^{-\frac{1}{2} \left\{ (X_1 - \mu_1)' (X_1 - \mu_1) + (X_2 - \mu_2)' (X_2 - \mu_2) + (\underline{\mu} - \underline{1})' (\underline{\mu} - \underline{1}) \right\}} d\underline{\mu}$$

$\Rightarrow$ the integrand would be a quadratic expression (on the power of exp and it will form a density of a BVN distribution after completing the square. So, due to Gaussian-Gaussian convolution, the final distribution is a 4-variate multivariate normal. We just need to find the mean and Covariance.

$$E(\underline{X}) = E E(\underline{X} | \underline{\mu}) = E \begin{pmatrix} \underline{\mu} \\ \underline{\mu} \end{pmatrix} = \begin{bmatrix} E(\underline{\mu}) \\ E(\underline{\mu}) \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$\text{Cov}(\underline{X} | \underline{\mu}) = I_4$$

$$\Rightarrow E \text{Cov}(\underline{X} | \underline{\mu}) = I_4$$

$$\text{Cov}(\tilde{x}) = E[\text{Cov}(\tilde{x}|\tilde{\mu})] + \text{Cov}[E(\tilde{x}|\tilde{\mu})]$$

$$= I_4 + \text{Cov}\left(\begin{matrix} \tilde{\mu} \\ \tilde{\mu} \end{matrix}\right)$$

$$= I_4 + \begin{bmatrix} I_2 & I_2 \\ I_2 & I_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} + \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 2 & 0 & 1 & 0 \\ 0 & 2 & 0 & 1 \\ 1 & 0 & 2 & 0 \\ 0 & 1 & 0 & 2 \end{bmatrix}$$

4/ We assume $f(x) = \sum_{\ell=1}^L \beta_{\ell} B_{\ell}(x)$. Thus, given β_{ℓ} 's, $f(\cdot)$ is fully known.

$$Y_i | \beta \sim \text{Normal}\left(\sum_{\ell=1}^L \beta_{\ell} B_{\ell}(x_i), \sigma^2\right)$$

$$\Rightarrow Y_i | \beta \sim N(\tilde{z}_i' \beta, \sigma^2)$$

$$\tilde{z}_i = [B_1(x_i), \dots, B_L(x_i)]'$$

$$\beta = (\beta_1, \dots, \beta_L)'$$

σ^2 is known.

$$L(\beta) \propto e^{-\frac{1}{2\sigma^2} \sum_{i=1}^n (Y_i - \tilde{z}_i' \beta)^2}$$

$$\propto e^{-\frac{1}{2\sigma^2} (Y - Z\beta)' (Y - Z\beta)}$$

$$\tilde{y} = \begin{pmatrix} Y_1 \\ \vdots \\ Y_n \end{pmatrix}, \tilde{Z} = \begin{bmatrix} \tilde{z}_1' \\ \tilde{z}_2' \\ \vdots \\ \tilde{z}_n' \end{bmatrix}$$

$$\ell(\beta) = c - \frac{1}{2\sigma^2} (Y - Z\beta)' (Y - Z\beta)$$

$$\frac{\partial}{\partial \beta} \ell(\beta) = -\frac{1}{2\sigma^2} (-2\tilde{Z}'\tilde{y} + 2\tilde{Z}'Z\beta)$$

$$\frac{\partial^2 \ell(\beta)}{\partial \beta^2} = -\frac{2\tilde{Z}'Z}{2\sigma^2}, \text{ independent of } \beta \text{ and } \tilde{y}$$

$$\Rightarrow E\left(-\frac{\partial^2 \ell(\beta)}{\partial \beta^2}\right) = -\frac{\partial^2 \ell(\beta)}{\partial \beta^2} = c \text{ (w.r.t } \beta)$$

$$\Rightarrow I(\beta) = c. \quad \Rightarrow \pi(\beta) \propto \sqrt{I(\beta)} \Rightarrow \pi(\beta) \propto 1.$$

$$5/: M_1: Y \sim N(0, \sigma^2) \Rightarrow L(Y|M_1) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2} Y^2}$$

$$M_2: Y/\mu \sim N(\mu, \sigma^2), \quad \mu \sim N(\mu_0, \sigma_0^2)$$

$$L(Y|M_2) = \int \underbrace{\pi(Y|\mu)}_{\text{Normal}} \underbrace{\pi(\mu)}_{\text{Normal}} d\mu. \quad \Rightarrow \text{Normal density.}$$

So, just find its mean and variance.

$$E(Y) = E[E(Y|\mu)] = E[\mu] = \mu_0$$

$$V(Y) = V[E(Y|\mu)] + E[V(Y|\mu)] = V(\mu) + E(\sigma^2) = \sigma_0^2 + \sigma^2$$

$$L(Y|M_2) = \frac{1}{\sqrt{2\pi(\sigma_0^2 + \sigma^2)}} \cdot e^{-\frac{1}{2(\sigma_0^2 + \sigma^2)} (Y - \mu_0)^2}$$

$$\frac{L(Y|M_2)}{L(Y|M_1)} = \sqrt{\frac{\sigma^2}{\sigma_0^2 + \sigma^2}} \cdot e^{-\frac{1}{2(\sigma_0^2 + \sigma^2)} (Y - \mu_0)^2 + \frac{1}{2\sigma^2} Y^2}.$$